

Sheet OS

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1. Explain the difference between Access Control Lists (ACLs) and Capability Lists (C-Lists). Provide one advantage and one disadvantage of each.

ACL (Access Control List): Stores the access rights with each object.

- **Advantage: Efficient for revocation — just modify the ACL of the object.**
- **Disadvantage: Confused deputy problem — it's hard to distinguish whether a request is due to the user or the program acting on the user's behalf.**

C-List (Capability List): Stores access rights with each domain (or user/process).

- **Advantage: Efficient for access checking — just check the process's C-List.**
- **Disadvantage: Costly for revocation — you need to search and modify C-lists of all users**

2. What is the Confused Deputy Problem? How can it be avoided when designing secure systems?

Definition: Arises when a program (deputy) that has higher privileges performs actions on behalf of a user without properly distinguishing their authority.

Solution: Design systems to clearly separate user rights from program rights; use C-Lists to reduce ambiguity

3. Describe the Bell-LaPadula Model and explain its two main properties. How does it ensure confidentiality?

Goal: Ensure confidentiality.

Properties:

- 1. Simple Security Property: No read-up.**

2. * (Star) Property:** No write-down.**

How it ensures confidentiality: Prevents users from reading data at higher classification levels and from writing to lower levels

4. What is Mandatory Access Control (MAC) and how is it different from Discretionary Access Control (DAC)? Provide a real-life example.

MAC (Mandatory Access Control): The system enforces access rules; users cannot override them.

- **Example: Military document classifications.**

DAC (Discretionary Access Control): Object owners define access.

- **Example: File permissions in Linux**

5. How do Protection Domains and the Access Matrix model relate to implementing least privilege and need-to-know principles?

Protection Domains: Define what resources a process/user can access.

- **Access Matrix: Maps domains to objects with rights.**
- **Relation to Principles:**
 - **Least Privilege: Only needed permissions are assigned.**
 - **Need-to-Know: Access limited to required info only.**

6. Compare and contrast static and dynamic domain associations. When is dynamic association preferred?

Static: Domain is fixed for process lifetime.

Dynamic: Process can switch domains during execution.

Preferred: Dynamic, to enforce least privilege at different execution stages.

- **7. Describe how an access matrix is maintained. What are the roles of the *, +, and owner rights?**

*** (Star): Copy right within same object.**

- **+ (Plus): Transfer right and delete original.**
- **Owner: Add/remove rights from object column.**

8. Explain how Capability Lists (C-Lists) are protected and give examples of how they are implemented securely in real systems.

Methods:

- 1. Tagged architecture (hardware protected).**
- 2. C-list inside OS (like file descriptors in UNIX).**
- 3. Cryptographic protection (used in distributed systems).**

Examples: UNIX file descriptors, remote server capabilities with cryptographic checks

9. Discuss the drawbacks of using biometrics for authentication. Why is fallback authentication important in biometric systems?

Expensive, needs special hardware.

High accuracy ⇒ false negatives.

Fallback important: Allows recovery (e.g., iPhone uses PIN).

Biometrics are non-revocable and unsuitable for remote authentication.

10. What are the main criteria used to compare biometric methods (e.g., fingerprints, iris)? Explain any three

Universality: Applies to all people?

Distinguishability: Can it differentiate users clearly?

Permanency: Is the biometric stable over time?

Mcqs

- 1- B
- 2- C
- 3- B
- 4- D
- 5- C
- 6- C
- 7- C
- 8- B
- 9- B
- 10- C

